A GENTLE GUIDE TO RESARCH METHOD (GNS 228)

HOW TO WRITE A RESEARCH PAPER

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RESEARCH METHODOLOGY

DEFINITION OF RESEARCH

The term research is like any other social science concept because it could be define differently by different authors and each will define the concept based on his perception and paradigm direction or level of proactivity. Ordinarily it could be inquiry into unknown which must start from known or process of finding out solution to in planed and systematic way. Some define research as a systematic attempt to provide answer to problems.

However there are many more definitions of the concept, but within the context of introduction we shall gear our definition toward solutions to problems or answers to question through planned and systematic approach. Ingrained in the above definitions is the fact that, they are tailored **"as a systematic activity that is essentially based on scientific procedure with the ultimate purpose of gathering data, information and facts for the advancement of knowledge."**

OBJECTIVES OF THE SCIENTIFIC RESEARCH PROCESS

- Capture contextual data and complexity
- Identify and learn from the collective experience of others from the field
- Identification, exploration, confirmation and advancing the theoretical concepts.
- Further improve educational design

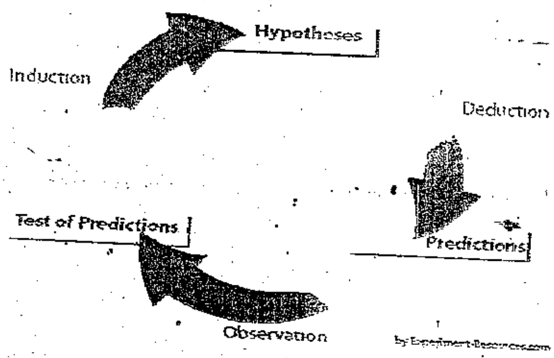
There are several important aspects to research methodology. This is a summary of the key concepts in scientific research and an attempt to erase some common misconceptions in science.

Steps of the scientific method are shaped like an hourglass - starting from general questions, narrowing down to focus on one specific aspect, and designing research where we can observe and analyze this aspect. At last, we conclude and generalize to the real world.

KEY CONCEPTS OF THE SCIENTIFIC METHOD

Researchers organize their research by formulating and defining a research problem. This helps them focus the research process so that they can draw conclusions reflecting the real world in the best possible way.

HYPOTHESIS



In research, a hypothesis is a suggested explanation of a phenomenon.

A null hypothesis is a hypothesis which a researcher tries to disprove. Normally, the null hypothesis represents the current view/explanation of an aspect of the world that the researcher wants to challenge.

Research methodology involves the researcher providing an alternative hypothesis, a research hypothesis, as an alternate way to explain the phenomenon.

The researcher tests the hypothesis to disprove the null hypothesis, not because he/she loves the research hypothesis, but because it would mean coming closer to finding an answer to a specific problem. The research hypothesis is often based on observations that evoke suspicion that the null hypothesis is not always correct.

In the Stanley Milgram Experiment, the a null hypothesis was that the personality determined whether a person would hurt another person, while the research hypothesis was that the role, instructions and orders were much more important in determining whether people would hurt others.

THEORY

Theory could be defined as an ideas, beliefs or claims about something which may not be necessarily true or base on reasoning.

Generally theories are principle on which a subject of study is based upon. For example a theory that wearing Hats makes men go bald, which may be true or false is subject to scientific exercise.

VARIABLES

A variable is something that changes. It changes according to different factors. **Random** variables change easily, like the stock-exchange value, while other variables are almost constant, like the name of someone. Researchers are often seeking to measure variables. The variable can be a number, a name, or anything where the value can change.

An example of a variable is temperature. The temperature varies according to other variable and factors. You can measure different temperature inside and outside. If it is a sunny day, chances are that the temperature will be higher than if it's cloudy. Another thing that can make the temperature change is whether something has been done to manipulate the temperature, like lighting a fire in the chimney.

In research, you typically define variables according to what you're measuring. The independent variable is the variable which the researcher would like to measure (the cause), while the dependent variable is the effect (or assumed effect), dependent on the independent variable. These variables are often stated in experimental research, in a hypothesis, e.g. "what is the effect of personality on helping behaviour?"

In explorative research methodology, e.g. in some qualitative research, the independent and the dependent variables might not be identified beforehand. They might not be stated because the researcher does not have a clear idea yet on what is really going on.

Confounding variables are variables with a significant effect on the dependent variable that the researcher failed to control or eliminate - sometimes because the researcher is not aware of the effect of the confounding variable. The key is to identify possible confounding variables and somehow try to eliminate or control them.

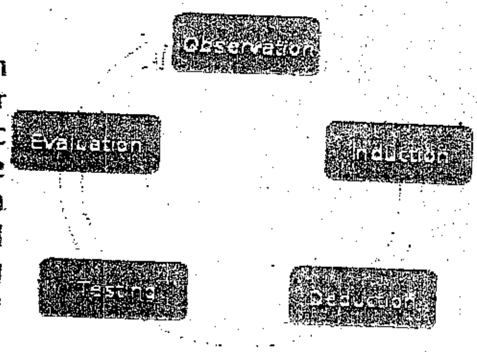
TYPE OF RESEARCH

The inquiry into unknown or the art and process of pushing the frontier of ignorance have over the years sank into the areas that little or nothing has ever been known or studied. On the basis of this, painstaking explanation shall be on the following most important types of research.

EMPIRICAL RESEARCH

Empirical Research can be defined as research based on experimentation or observation (evidence). Such research is conducted to test a hypothesis.
By Dr. Hani (2009)

The word empirical means information gained by experience, observation, or experiment. The central theme in scientific method is that all evidence must be empirical, which means it is based on evidence. In scientific method the word "empirical" refers to the use of working hypothesis that can be tested using observation and experiment.



Empirical data is produced by experiment and observation.

OBJECTIVES OF THE EMPIRICAL RESEARCH

- Go beyond simply reporting observations
- Promote environment for improved understanding
- Combine extensive research with detailed case study
- Prove relevancy of theory by working in a real world environment (context)

REASONS FOR USING EMPIRICAL RESEARCH METHODS

- Traditional or superstition knowledge has been trusted for too long
- Empirical Research methods help integrating research and practice
- Educational process or instructional science needs to progress

ADVANTAGES OF EMPIRICAL METHODS

- Understand and respond more appropriately to dynamics of situations
- Provide respect to contextual differences
- Help to build upon what is already known
- Provide opportunity to meet standards of professional research

In real case scenario, the collection of evidence to prove or counter any theory involves planned research designs in order to collect empirical data. Several types of designs have been suggested and used by researchers. Also accurate analysis of data using standard statistical methods remains critical in order to determine legitimacy of empirical research.

Various statistical formulas such uncertainty coefficient, regression, t-test, chi-square and different types of ANOVA (analysis of variance) have been extensively used tp form logical and valid conclusion.

However, it is important to remember that any of these statistical formulas don't produce proof and can only *support* a hypothesis, *reject* it, or do neither.

EXPERIMENTAL RESEARCH

Experimental research is commonly used in sciences such as sociology and psychology, physics, chemistry, biology and medicine etc. By Experiment-Resources.com (2008)

It is a collection of research designs which use manipulation and controlled testing to understand causal processes. Generally, one or more variables are manipulated to determine their effect on a dependent variable.

Method The experimental is a systematic and scientific approach to research in which the researcher manipulates one or more variables, and controls and measures any change in other variables.

Experimental Research is often used where:

1. There is time priority in a causal relationship (cause precedes effect)
2. There is consistency in a causal relationship (a cause will always lead to the same effect)
3. The magnitude of the correlation is great.

The word experimental research has a range of definitions. In the strict sense, experimental research is what we call a true experiment.

This is an experiment where the researchers manipulates one variable, and control/randomizes the rest of the variables. It has a control group, the subjects have been randomly assigned between the groups, and the researcher only tests one effect at a time. It is also important to know what variable(s) you want to test and measure.

A very wide definition of experimental research, or a quasi experiment, is research where the scientist actively influences something to observe the consequences. Most experiments tend to fall in between the strict and the wide definition.

A rule of thumb is that physical sciences, such as physics, chemistry and geology tend to define experiments more narrowly than social sciences, such as sociology and psychology, which conduct experiments closer to the wider definition.

AIMS OF EXPERIMENTAL RESEARCH

Experiments are conducted to be able to predict phenomenon. Typically, an experiment is constructed to be able to explain some kind of causation. Experimental research is important to society - it helps us to improve our everyday lives.

CONDUCTING THE EXPERIMENT

An experiment is typically carried out by manipulating a variable, called the independent variable, affecting the experimental group. The effect that the researcher is interested in, the dependent variable(s) is measured.

Identifying and controlling, non-experimental factors which the researcher does not want to influence the effects, is crucial to drawing a, valid conclusion. This is often done by controlling variables, if possible, or randomizing variables to minimize effects that can be traced back to third variables. Researchers only want to measure the effect of the independent variable(s) when conducting an experiment, allowing them to conclude that this was the reason for the effect.

EXPLANTORY RESEARCH (why)

As the name implies, this type of research deals with the political or social problems. It seeks to address the question why certain problems exist. For example, why one child is more intelligent then the other, why students performance is reducing unlike hitherto. These are all cases that demand more explanation on why the issue occurred.

FACT FINDING (what happened)

These are research cases with a base or known case without adequate information to ascertain its truthfulness. Information needed here will be for more or adequate description of the case at hand. In this case, the researchers mind will be on answering the question "what happened" as a basis for understanding the case at hand. For example what do people in Mubi community think about drugs abuse and nasty dressing in ADSU and Poly. And how does it affect the original culture of the populace.

In these types of cases, the researcher seeks to find out more information on the existing case (drugs abuse and dressing). However in the case of carrying explanatory or fact finding research, researchers run into problems. In an attempt to solve the problems, various research methods such as **Historical, Descriptive,** and **Correlation** research method are adopted as a solution.

Descriptive Designs

Aim: Observe and Describe

- Descriptive Research
- Case Study
- Naturalistic Observation
- Survey (The Questionnaire is also a technique used in many types of research designs)

While historical research are curious about what happen in the past so that it can provide guidance on how to improve the future, the descriptive research are curious about what is happening currently so that it could help to predict the better outcome of the condition. Also the correlation method investigates one event with the intention of making a prediction to another event. All predictive studies are therefore correlational.

CHOOSING THE RESEARCH METHOD

The selection of the research method is crucial for what conclusions you can make about a phenomenon. It affects what you can say about the cause and factors influencing the phenomenon.

It is also important to choose a research method which is within the limits of what the researcher can do. Time, money, feasibility, ethics and availability to measure the phenomenon correctly are examples of issues constraining the research.

CHOOSING THE MEASUREMENT

Choosing the scientific measurements are also crucial for getting the correct conclusion. Some measurements might not reflect the real world, because they do not measure the phenomenon as it should.

SIGNIFICANCE TEST

To test a hypothesis, quantitative research uses significance tests to determine which hypothesis is right.

The *significance test* can show whether the null hypothesis is more likely correct than the research hypothesis. Research methodology in a number of areas like social sciences depends heavily on significance test.

A significance test may even drive the research process in a whole new direction, based on the findings.

The t-test (also called the Student's T-Test) is one of many statistical significance tests, which compares two supposedly equal sets of data to see if they really are alike or not. The t-test helps the researcher conclude whether a hypothesis is supported or not.

DRAWING CONCLUSIONS

Drawing a conclusion is based on several factors of the research process, not just because the researcher got the expected result. It has to be based on the validity and reliability of the measurement; how good the measurement was to reflect the real world and what more could have affected the results.

[illegible] empirical evidence and the logic/thinking leads to the conclusions. Anyone should be able to check the observation and logic, to see if they also reach the same conclusions.

Errors of the observations may stem from measurement-problems, misinterpretations, unlikely random events etc.

A common error is to think that correlation implies a causal relationship. This is not necessarily true.

GENERALIZATION

Generalization is to which extent the research and the conclusions of the research apply to the real world. It is not always so that good research will reflect the real world, since we can only measure a small portion of the population at a time.

INTRODUCTION TO PROBABILITY

The understanding of probability principle is central to statistical inference and also has essential information for the full comprehension of statistical method and procedures. Probability is defined as the chance of occurrence of some elementary events. It is the ratio of what you obtained compared to all that is available or possible. For example, if we take the probability of event to be the number of trials and the particular event occurred '**a**' times in '**n**' trials then probability of occurrence of **a=a/n**. This means that probability ranges from 0-1; that is $0 \leq p(x) \geq 1$ probability of an event x occurring (p) and probability of not occurring (q) can be represented as $p+q=1$.

At zero (0) the event presumably cannot occur but if it is one (1) then it is a certainty. If for example, we toss a coin, it can either fall head (H) or tail (T). Head and tail are the elementary event of tossing a coin. Also, if we throw a die, we can obtain 1, 2, 3 ...or 6. Thus, 1, 2, 3, or 6 are the elementary event in throwing a die. Some other important definition of probability terminologies are;

SAMPLE SPACE (S)

This is all the outcomes that are possibly in the event, e.g. in tossing a coin $S = H \& T$ while in throwing a die $S = 1 \cdots 6$

MUTUALLY EXCLUSIVE EVENTS

Two events, are mutually exclusive if the outcome of one of them does not affect the occurrence or non occurrence of the other. For example, the probability of GNS 228 Class holding today and the probability of Mr B. Marrying a second wife are mutually exclusive. Then, their intersection i.e $A \cap B = 0$ (empty set). E.g. if A=set of even numbers from 1.....10 i.e. 2, 4, 6, 8, 10 and B= set of odd numbers from 1....10 i.e. 1, 3, 5, 7, 9, then $A \cap B = 0$ (empty set).

COMPLEMENTARY EVENTS

Two events are complementary if their probabilities add up to 1 e.g. $P(A)+P(B)=1$. Hence $P(A) = 1 - P(B)$

DISJOINT EVENTS

Two events A and B are said to be disjoin i.e. $A \cap B = \emptyset$

Example (A)

A die is thrown. Find the probability of

1- Obtaining odd numbers
2- Even numbers
3- Numbers greater than 4
4- Numbers greater than 6

SOLUTINS

S= 1, 2, 3, 4, 5, 6

1- P(odd no) = (1, 3, 5)= 3/6=1/2 =0.5
2- P(even no) = (2, 4, 6)= 3/6 = 1/2 = 0.5
3- P(no> 4)= (5, 6)= 2/6= 1/3= 0.33
4- P(no> 6) =(∅)

EXAMPLE (B)

A ball is drowned at random from a bag containing 10 red, 30 white and 20 blue balls. Find the probability that a ball drowned at random is;

1- Red ball
2- White ball
3- blue ball
4- Orange ball.

SOLUTION

Total=10 red + 30 white + 20 blues= 60 balls

1- P(red)= 10/60=0.17
2- P(white)= 30/60= ½= 0.5
3- P(blue)= 20/60= 1/3= 0.33
4- P(orange)=∅

ADDITIONAL LAW

If events A, B, C, are mutually exclusive, the probability of A or B or C or.... Happening is the sum of their individuals probabilities; p (A) + p(b) + p(C) +......

EXAMPLE (C)

A number is chosen at random from the set (2, 4, 6...18, 20). Find the probability of either a factor of 18 or a multiplier of 5?
∈= (2,4 ... 18,20)

F= (factors of 18) = (2, 6, 18),

M = (multiple of 5) = (10, 20)
F and M are mutually exclusive.
P (F) + P (M)

P (F) = = 3/10

P (M) = =2/10

P (F) + P (M) + 3/10 +2/10=5/10
Note additional law use and or, or either or.

PRODUCT LAW

For independent event their probabilities of happening is the product of their individual probabilities. $P(A)\cdot P(B)\cdot P(C)$

EXAMPLE (D)

Five (5) Girls name and three (3) boys' names are in a box. 2 names are picked at random one after the other without replacement. What is the probability that both names are girls?

G = 5, B = 3 $\Sigma = 8$

$P(G1)\cdot p(G2) = p(\frac{5}{8})\cdot p(\frac{4}{7}) = \frac{5}{8}\times\frac{4}{7} = \frac{5}{14}$

SAMPLE

Sample can be defined as a proportion of a population selected for observation and analysis with a view to making inferences about the population. In selecting a sample the investigator must ensure that he is not biased. In other words, he should select his sample at random, that is given every subject equal and independent chances of being selected for the study. A sample is usually fractional part of the population. It is ideally a representative or replica of a population. Sampling therefore is taken any portion of a population or universal representative of that population. And it could be of any object living or non living depends on the type of study at hand.

TYPE OF SAMPLNG

There are two major types of sampling

PROBABILITY SAMPLING

Probability sampling involves the following;

(A) RANDOM SAMPLING:

Random sampling is the type of sapling method of drawing a portion of the population so that each member of the population has an equal chance of being selected. Simple random sampling is a correct example of balloting. Random sampling is the process of selecting a sample in such a way that all individual on the defined population have an equal and independent chance of being selected for the sample. E.g. take a pick lottery method.

(B) SYSTEMATIC SAMPLING

This is appropriate where a population has been listed accurately. The first or fourth name is selected randomly. Systematic sampling, this procedure involves drawing a sample by taking every kth case from a list of the population. Thus the major difference between systematic sapling and other type of sampling is the fact that all number of the population does not have an independent chance of being selected for the sample. Once K is determined as the first Kth element decided then, the rest of the individuals to be included in the sample are automatically determined. To use this procedure, you need to decide the number of object in your sample (n). The already known total number of individual in the population is N. In other to determine K, the sample interval, you divide N with n i.e. K=N/n the first number of the randomly selected from the first K number of the list and their every Kth number of the population is selected for the sample. For example, a population of 500 subjects and a samples study (n) of 50. K is given by 500/50=10. You can then select any of the first 10 cases, on the top of the list then add 10+5 to pick 15th. Then, add 10 and pick 25 case etc until you get the require sample study of 50.

(C) STRATIFIED RANDOM SAMPLING.

This is applied where the population is made of smaller homogeneous groups that need to be represented in the study in order to get more accurate information. E.g. where a population is made up of individuals with different socio-economic status and other characteristics as sex, age, academic qualification etc there is need to sub-divide the population in to smaller groups as such characteristics will constitute a source of differences in opinion, attitude or perception of population. It is the process of selecting a sample in such a way that it identifies sub proportion that exist in the population. i.e selection of individual in a sample from the subgroups in proportion to the study of each subgroup in the population. The major advantage of stratified sampling is that of guaranteed representation of defined group in the population.

(D) CLUSTER SAMPLING

As the name implies, the population is large and widely scattered so that investigator & convenient to select cluster or group at random and all the members studied. It is impracticable to obtain or compile a list of all the numbers of a population say engineers or buildings or cars in Nigeria. The cost of studying subject of a population that spans a very wide geographical area is prohibitive. All the members of the selected groups have the same characteristics. Any intact groups of similar characteristics are a cluster e.g. classrooms, schools, city blocks car machines etc, the procedural requirement for cluster sampling are that;

- The cluster include in a study must be randomly selected from the population of cluster.
- Once a cluster is selected then all the members of the cluster must be included in the sample.

NON PROBABILITY (biased sampling)

This is the method of sampling in which the chances of an element or person being included in the sample is not the same. Some of these methods are:

(A) Quota sampling.

This is a form of sampling to ensure that all the subset of the population are represented in these proportion especially in market or research surveys. The requirement in quota sampling is not that the various population strata be sampled in their correct proportion but, rather that there be enough cases of each stratum to make possible an estimate of the population value.

(B) Purposive sampling.

This is the sampling method whereby the research just pick those considered to poses the required attributes or information. It is also called expert or judgement sampling. This is sampling by conveniences. E.g. where you picked and studied building along the major roads as only these are easily accessible. In this case, ensure that you make clear the reason for using this sampling technique which is unscientific.

(C) Accidental Sampling

This implies choosing any member of the population that is available at the given time until the desired number is achieved. Participation here is based on availability and willingness. Pressman, Journalist, Astronomies, Pharmacists, Medical practitioners are some of the fields where representatives is not the driving force. The precision of This method is not guaranteed. The result may not be generalised into future consideration.

ADVANTAGESN OF PROBABILITY SAMPLES

1- This method is simple, purposeful and methodical.
2- The method uses rules of probability which means the result or outcome can be ascertain easily. –
3- It guaranteed representation of defines group in population.
4- It logical, definite and desired sample size of population.
5- The method is scientific in nature that is why most of the result are exact or near exact.

DISADVANTAGES

1- It requires the knowledge of probability and statistics.
2- The method is laborious and the mixing may not be enough.
3- It's most tedious and time consuming.
4- It emphasis is on the types of differences among the strata, equal number of cases will be selected of a population that spans every wide geographical areas is prohibitive.

ADVANTAGE OF NON PROBABILITY SAMPLES

1- It is a choice or judgement
2- The research will just pick or consider information that is closed to it.
3- Participation is based on availability or willingness.
4- It does not involve much knowledge, time and money in using such types of method.
5- There is here is that error of judgement in the selection will tend to counter balance each other.

DISADVANTAGES OF NON PROBABILITY SAMPLING

1- It is unscientific therefore cannot be considered for generalization.
2- The result may not be generalised in to future considerations.

3- Most of the strata sample are not correct and cannot be enough to make possible estimate of the population.
4- It is weakest sample therefore it cannot be essential to avoid it used.

USES OF SAMPLING IN THE FIELD OF ENGENEERING

1- It provides and processed data more feasible when the volume of work is given e.g. evaluation of buildings and modern farming tools in North East Zone.
2- It makes it possible to conduct large-scale studies at greater speed.
3- Engineering as a field of disciplines involve the use of huge amount of money to carry out a research with use of sampling the cost will be reduced.
4- It sampling if applied with highly trained person or specialised equipment will cover wide scope.
5- With the use of sampling in the field of engineering, sampling techniques makes it possible the conducting of otherwise impossible studies by selecting representatives units from the population.

STEPS IN THE SAMPLING PROCESS

A- Identify the population
B- Obtain a list of unit in the population
C- Determine the size of the sample so that all characteristics of the population are presented
D- Drawing a unit from the list so they are representative of the total universe.

DATA COLLECTION

Data can be defined as raw information, fact or statistics used for references or analysis.

TYPES OF DATA

➢ **Primary data**

The primary data are information sources which contain original accounts of events, report or observation given by someone that actually observed or witnessed and reported the event or phenomenon. Example of such data is research project reports, theses or dissertations, journals, abstracts, publications, conferences proceedings, technical reports.

- **Secondary Data**

The secondary data are sources which contain the account of an event or phenomenon by someone who did not actually witnessed or observes the event or phenomenon. E.g. are textbooks, other books review of research report, book reviews, encyclopaedias, dictionaries, Almanac, international organization material published etc.

DATA COLLECTION

The data must be collected carefully to ensure its quality. Biased and error must be avoided. Again, there is the need to plan before data collection is connected. Haw the data will be organised and presented should have been determined, the use of tables, figures and charts are essential in organizing and summarizing data. Here the investor should specify the method or instrument used for data collection. The data required to provide answer to research question or test hypothesis are collected with the aid of research instrument. It is important here to show how the data is gathered. Whether the data was gathered personally or through the involvement of research assistant or it was collected through the use of mail system or telephone.

VALIDATION OF INSTRUMENT

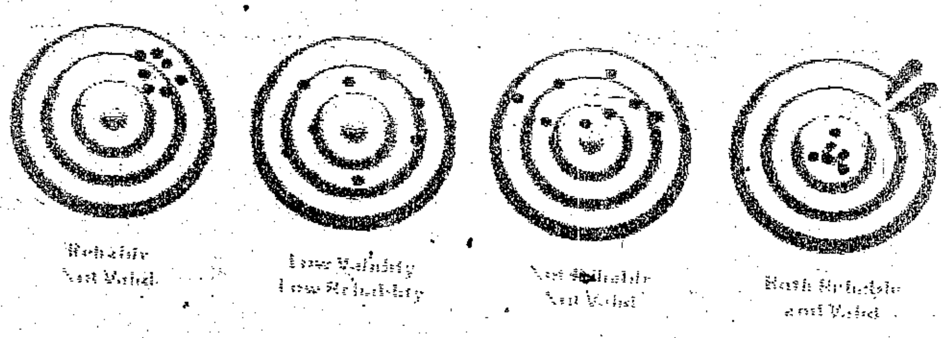
It is important to validate the instrument designed for data collection to ensure it measure what is design to measure. It is necessary that researchers should validate the instrument designed for data collection. The validation is usually done by expert in the field who ensure on face value that the.instrument is appropriate in measuring what is designed to measure.
After the validation of instrument, it is pilot tested on a similar but smaller sample. The essence is to find out how the respondents will react to the instrument.

RELIABILITY OF THE INSTRUMENT

The researcher must report.in details how to establish the reliability of the instrument. The reliability of an instrument or a test of degree to which a test or an instrument is consistent in measuring whatever it designed to measure. In other word it is the degree to which the test or the instrument measures the same time after time and item after item. It is also the solubility dependability of the research instrument cannot be subjectively determined, but rather be computed from some data. The index of reliability is usually expressed as a coefficient reflecting the extent to which a test is free of error variance. There are many techniques for determining the reliability of research instrument.

VALIDITY AND RELIABILITY

Validity refers to what degree the research reflects the given research problem, while Reliability refers to how consistent a set of measurements are.



by Experiment-Resources.com

Reliability may be defined as "Yielding the same or compatible results in different clinical experiments or statistical trials" (the free dictionary). Research methodology lacking reliability cannot be trusted. Replication studies are a way to test reliability.

Both validity and reliability are important aspects of the research methodology to get better explanations of the world.

ADMINISTRATION OF INSTRUMENT

The researcher should describe briefly how the instrument was administered on the respondents, whether personally or by mail in the case of questionnaire, the researcher should specify the number of copies of the questionnaire distributed and the number returned hence the percentage retune.

DATA ANALYSIS

The type of research used for the study is a pointer to the statistical techniques that could be used. This also depends on the type of hypothesis and the type of data (normal, ordinal interval, or ratio).

The researcher specifies the statistical tools he/she employed for data analysis. He must ensure that he employs appropriate statistical tools in other to get valid results. Two types of statistical tools are at the disposal of the researcher for data analysis. The descriptive statistics and the inferential statistics.- If the research is concern with ensuring research question, he should use descriptive statistics such as proportion/percentage mean, median, mode, standard deviation, and variety of procedures for graphical illustration such as charts, bar graphs, histograms, and frequency polygons.

If the researcher is concerned with testing hypothesis, then he should use inferential statistics such as chi-square ( ), analysis of variance (ANOVAs), analysis of covariance and t-test etc.

OBSERVATION

An observation is the act of watching or supervising an experiment, event, activity or an occasion and taking record of the happiness, such way are also used in research for gathering data. It can be personal or by proxy as the case may, i.e. participatory or non participatory, by participating observation, the observer takes part in action while in non-participatory he may assign someone to stand close by and watches what is taking place, then he record the information, fact or the event for his research. Observed data usually gain validity if the observer in unbiased, and they are referred to be primary data.

ADVANTAGES OF OBSERVATION

1- The data are purposeful and directly useful. Since they were determined before the event or action.
2- The data generated are fast and short this is because records are made as the occurred during the event.
3- This is cheaper, easy depending on nature of experiment or activity concerned.

DISADVANTAGE OF OBSERVATION

1- It may lack occurrences and some area of event may not be covered during the observation.
2- Observation is subject to personal whims which may assume to be inconsistent in recording of events.
3- Observers may be bias or unbiased as human beings are dynamic in nature.

QUESTIONAIRE

Questionnaire is one of the research instruments that are used to collect basic descriptive information from a large sample. It consists of a set of question designed to gather data or information to be used to answer the research question and or test relevance research hypothesis for the study.

TYPES OF QUESTIONNAIRE

There are mainly two types of questionnaire, it can be used alone or combined as the cases may be.

a- **Structured** or fixed response questionnaire
b- **Unstructured** or free response questionnaire.

STRUCTURED QUETIONNAIRE

In structured questionnaire the respondent is restricted to some response options. The respondent is required to select or check on item from list of suggestion responses that best suit his/her response. Sometimes however, anticipated response is provided to permit the respondent to give response that had not been anticipated by the researcher.

UNSTRUCTURED QUESTIONNAIRE

This is the type which if appropriately employed where the researcher cannot predict what the respondent responses are likely to be. It allows the respondent to answer the question full and freely in their own words. Unstructured questionnaire not provide any response option or clues for the respondents.

ADVANTAGES OF A QUESTINNAIRE

1- It is cheaper and more affordable and both the researcher and respondent.
2- It eliminates personal contacts and related influence on the respondent and the data.
3- It facilitate data analysis, i.e. it is easy tabulate and analyse
4- It is less expensive and also less time consuming.

LIMITATION OF QUESTIONAIRE

1- There may be biases which tend to influence the types of question asked.
2- Respondent that returns the questionnaire may not constitute a representative sample.
3- Response to question may tend to give answer contrary to ral opinion of the respondent.

DISADVANTAGE OF UNSTRUCTURED QUESTIONNAIRE

1- Respondent may avoid answering some question to which they have no clues or may give answers.
2- Communication gap due to level of illiteracy may deny self expression in terms of data collection.
3- Difficulties may arise in classifying and quantifying responses.
4- Analysis and summarising different responses may be difficult and time consuming

QUALITIES OF A QUESTIONNAIRE

1- RELEVANCE; it should be relevant to purpose of research
2- CONSISTENCY; it should yield consistent responses of the
3- USABILITY; a good questionnaire should be such that it is usable
4- CLEARITY; the items and the instruction should be clear enough to avoid possible misinterpretations.
5- QUANTIFIABILITY; it should be easy to assign numerical values to the response in a manner that is systematic.
6- LEGIBILITY; a good questionnaire should be legible.

HOW TO DESIGN A QUESTIONNAIRE

Construction of questionnaire items usually follows a clearly defined procedure. The following steps are pertinent and are therefore suggest in developing questionnaire.

- Identify the purpose of the study the questionnaire is expected to accomplish.
- Obtain and review similar instrument and literature.
- Develop/write and items pool which elicit the desired information from the subject considering all factors such as
  - Characteristics of the respondents, avoiding leading questions.
  - The type of questionnaire format, destruction between what ought to be what is.

- The length of the questionnaire, avoid unnecessary presumption in writing question. Avoid embarrassing question which may hurt the respondent and doubled barrelled question.
- Subject the questionnaire to validation to validate process by panel of expert to determine if the items concerned are legible.
- Establish the reliability of questionnaire.

ADMINISTRATION OF QUESTIONNAIRE

The researcher should describe briefly how the instrument was administered on the respondent whether personally or by mail. In the case of questionnaire, the researcher should specify the number of copies of the questionnaire distributed and the number returned, hence the percentage returned.

Administration of questionnaire to respondent is usually done when validity and reliability of the questionnaire are done. The research should ask the right question, phrase in the least ambiguous way.

INTERVIEW

Interview is the act of asking a person or group of person's question to which they provide answer there and then. Interview is an ordinary discussion and seeks to gather more information about a particular something. Interview is to collect information in order to predict how successful can the researcher addressed expected without undue delay.

TYPES OF INTERVIEW

Direct and indirect interview; direct interview occurred when the interviewer come into contact with the interviewer i.e. face to face. In an indirect interview a third party is involve or some time via internet, telephone or television as the case may be.

- **GROUP INTERVIEW**

Group interview as the name implies it requires four to six peoples to inter an interview room in order to be interviewed by the interviewer.

- **PROBLEM SOLVING INTERVIEW**

The interviewer always makes hypothetic assumption and asks the people comments. It is some time called situational interview.

- **PANEL INTERVIEW**

In such types of interview always one person appears in the member of interviewers.

ADVANTAGES OF INTERVIEW

- ✓ It provides a personal contact between the interviewers and the interviewees thereby enriching the information generated.
- ✓ Information obtained is usually more coherent, relevant and thus more reliable.
- ✓ There is prompt response to question which eliminate time consumption.
- ✓ It does not involve high cost as regard to questionnaire method.
- ✓ It provides an avenue to examine some before given answers to the situation at hand.

DISADVANTAGES OF INTERVIEW

- ✓ Interview might be very experience and not doing so might affect the research findings sufficiently negative.
- ✓ When interviewers have made up their mind very early i the interview, their behaviour betrays their decision which will affect the data or information.
- ✓ Interviewers decide to accept or reject within first three or four minute of the interview and then spend the remaining of the interview seeking evidence to confirm that first impression was right.
- ✓ Interview normally last for very short period of time before people can sustain an artificial behaviour for that period which may lead to affect the data to be collected.
- ✓ It is difficult to say that interview is a good measure for data collection in terms of sampling of data.

SCORING OF THE INSTRUMENT

This section could be integrated in to next method of data analysis, as it merely concerned with how the data was organise, this is indicate that the frequencies of the respondent where worked out and the percentages calculated. Where rating scale are used. Indicate the rating example as four-point scale and Likert-type scale.

(A) Strongly agree, Agree, Disagree, and Strongly Disagree.
(B) Very satisfied, Satisfied, Moderately Satisfied, Dissatisfied
(C) Very adequate, Adequate, Minimally adequate, Inadequate.
(D) Very important, Important, Somewhat important, Not important.
(E) Outstanding, Very good, Good, Fair, Poor.

Where rating scales are used under this section the numerical values assigned to the rating must be indicated. E.g.

- Strongly Agree 4
- Agree 3
- Disagree 2
- Strongly Disagree 1

Mentioning must be made at this stage of any special feature involved in scoring the responses. E. G rating the following statement on your attitude of well water for human consumption along the rating scales- Strongly Agree; Agree, Disagree, Strongly Disagree. Figure 1 below will connotes a sample questionnaire.

Figure - 1

Rating				
Statement	Strongly agree	Agree	Disagree	Strongly disagree
I enjoy drinking well water				
The water response to Omoar Soap when washing				

1- Mailing.
2- Administering it personally

MAILING

In administering mailed questionnaire, the researcher firs identifies the respondent to be involve in the study and then mail a copy of the questionnaire to each of them. The mail questionnaire should be accompanied with letter of transmittal to win the support or cooperation of the respondents so that they will complete and return of the questionnaire accordingly.

ADVANTAGES OF MAILED QUESTIONNAIRE

1- It is relatively cheap.
2- Information could be collected from many respondents in distant place at very low cost.
3- It gives respondent' the opportunity to complete the questionnaire at their convenient and in privacy.

DISADVANTAGES OF MAILED QUESTIONNAIRE

1- It has very low percentage of questionnaire return which limited the generalization of the result of the study.
2- The researcher can never be sure of the actual person who response to the questionnaire.
3- There is the likely hood of misinterpretation of some question because the researcher is not available on the spot to explain certain points to the respondent.

PERSNAL QUESTIONNAIRE ADMINISTRATION

The personal questionnaire administration is more common than the mailed questionnaire. The questionnaire is administered face to face to the respondent by either the researcher to research assistant.

ADVANTAGES OF PERSONAL ADMINISTRATION OF QUESTIONNAIRE

1- The percentage of return is very high.
2- The researcher can explain any point to the respondent.
3- It ensures that the actual individual for whom the questionnaire is meant is indeed the one who completes it.

DISADVANTAGES

1- It is time consuming and expensive particularly if subject are distributed over a wide area.
2- Travelling to all location to administer the questionnaire.
3- Intrusion of the researcher, that is the respondents has no privacy and is not sure of anonymity in responding to the question.

HOW TO WRITE A RESEARCH PAPER

This guide aims to help you write a research paper.

It contains an overview on writing academic papers such as the term paper, thesis, research paper or other academic essays written in the format of the research paper.

GENERAL OVERVIEW

Many students wonder about the writing process itself. The outline of the academic paper is very similar for most branches of science.

Creating an extended outline may help structure your thoughts, especially for longer papers. Here are a few samples outlines for research papers.

When writing a scientific paper, you will need to adjust to the academic format. The APA writing style is one example of an academic standard frequently used.

By the way, here is another great resource on how to write a research paper.

PREPARING TO WRITE A RESEARCH PAPER

Usually, the purpose of a research paper is known before writing it. It can formulate as a research paper question, a thesis statement or a hypothesis statement.

If you do not know what to write about, you will have to look for ideas for research paper topics.

STRUCTURE OF A RESEARCH PAPER

The structure of a research paper might seem quite stiff, but it serves a purpose: It will help find information you are looking for easily and also help structure your thoughts and communication.

Here is an example of a research paper. Here's another sample research paper.

PRILIMINARIES SECTIONS

The preliminary section of a research report comes before the actual body of the work and is paged in Roman numerals. The areas in order of occurrences as follows:

I. Cover page
II. Title page
III. Approval page
IV. Certification page
V. Acknowledgement page
VI. Abstract
VII. Table of content
VIII. List of tables
IX. List of figure

COVER PAGE:

The first display of the title of research is at the front cover of the report. E.g.

Figure- 2

CONSTRUCTION OF WELL WATER FOR HUMAN CONSUMPTION IN YOUR LOCALITY

BY

GOOGLUCK EBELE JONATHAN

GCFR/NG/ND/M/AG/2011

FEDERAL DEPARTMENT OF HUMAN DEVELOPMENT AND SOCIAL WELFAIRE FEDERAL POLYTECHNIC MUBI

SEPTEMBER 2011

Cover page of a report.

The technicalities to not in typing of this cover are
1- the space at the bottom and top of the prints marked
2- the typing should be centred between right and left

3- the other spaces should be then equally spaced
4- the typing for each section where more than a line

Apart from presentation of the title on front page, in the preliminary pages, the leaf is the leaf after the cover and which has no front.

Figure- 3

Just add this to figure 2 above

IN PARTIAL FULFILMENT OF THE REQUIREMENT FOR THE AWARD OF NATIONAL DIPLOMA IN......... ENGINEERING

BY

SEPTEMBER 2011

Title page of research report

The content of this title cover are the title of the work. The author and his/her identity number, the department, the institution and the date of completion of the programme.

The title of the research report should be concise and should summarise the main idea of the study.

A good title refers to the principal variable of theoretical frame work under investigation. E.g. an investigation is interested in finding out what children like a story book. The major function of the title is to inform reader about the study. The title should be self explanatory.

APPROVAL AND CERTIFICATION PAGES

Either approval page or certification page is used depending on the requirement of the institution. E.g.

Figure 4

APPROVAL PAGE	
This project has been approved for department of department of... engineering, Federal Polytechnic Mubi	
By	
Supervisor	External Examiner
External Examiner	Head of Department
Dean of Faculty	

ACKNOWLEDGEMENT

The acknowledgement should contain expression of appreciation of person who gave unusual assistance to the research in the project. It usually reported in third person singular. It is supposed to be academic in outlook and not a value for expression of personal emotions.

DEDICATION

This is optional. This could be included if the candidate wishes.

ABSTRACT

An abstract is a brief summary of the content of project, dissertation/thesis or paper presentation. The principal purpose of an abstract is to help the reader comprehend the content of the work quickly.

In writing an abstract, the student ensures that he does not include any information that does not appear in the main body of his study.

Abstract should be very brief but self contained and fully intelligible. A good abstract contains the followings statement of the problem, method of findings, and conclusion. Also an abstract should not be paragraph.

TABLE OF CONTENTS

This section provides an outline of the content and sub content areas of the report and the pages they appear. The content should include the chapter titles as well as the major sub- heading with each chapter.

LIST OF ILLUSTRATION

These are list of maps, pictures, figures etc where necessary in the report writing. It should be indicated after the table of contents.

NOTE OF GUIDELINE

Give the number of the table or figure and then the full title and the page number as shown in the project.

PART 2

MAIN BODY OF THE WORK

The main body of a project report come under the following headings;

CHAPTER ONE: introduction

CHAPTER TWO: literature review

CHAPTER THREE: research methodology and procedure

CHAPTER FOUR: result presentation and analysis of data

CHAPTER FIVE: discussion of findings, recommendation and conclusions.

ADDITIONAL PARTS FOR SOME ACADEMIC PAPERS

The following parts *may* be acceptable to include in some scientific standards, but may be inappropriate for other standards.

- Firm up the initial project idea, making sure that it tackles an interesting question in a way which will give you interesting outcomes whatever you find.
- Take advice about scoping the project – better to do something smaller well than something bigger badly.
- Think about the follow-on from the project (publication, links with potential employers, further degree, etc); if there isn't an obvious follow-on, then this suggests that you could design a better project.
- Do the detailed literature search for this topic – identify references for the underlying problem, for the seminal article, for milestone articles, and for a foundational article.
- Plan the project, working backwards from the hand-in date, and building in contingency time (spare time in case something goes wrong).
- Meet your supervisor regularly; take the initiative, and listen to their advice.
- Write up some sections early on, so you can get feedback on writing style.
- Think explicitly about what indicators of excellence you can build into every stage of your project.
- Show drafts to your supervisor and learn from their feedback.
- Be meticulous about presentation, particularly the formatting of your references section.
- Show the final draft to your supervisor well before the hand-in date.
- Be patient; remember that the more you put in, the more you get out; keep a sense of perspective and don't overwork. Better to work smart than to work hard but pointlessly.